

Standardized, Reproducible Parameter Curation for QSP Models

A Template-Based LLM Workflow

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Overview

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The Problem

The QSP Reproducibility Crisis

Current State of Parameter Curation

QSP models require **100+ parameters** from diverse literature, but:

- **Ad-hoc extraction**: No standardized methodology
- **Poor documentation**: Values cited without context or uncertainty
- **Propagation errors**: Parameters copied across models without verification
- **"Telephone game" effect**: Loss of biological context and accumulating errors
- **Difficult to verify**: Reviewers can't easily trace parameters to sources

Result

Models are difficult to reproduce, validate, and reuse

Why Standardization is Hard

What we need:

- Numerical values + uncertainty
- Biological context
- Experimental conditions
- Statistical distributions
- Literature provenance
- Cross-study synthesis

The barriers:

- Manual curation at this level can take **weeks to months**
- Adding documentation increases burden
- No widely adopted standards
- Trade-off: speed vs. thoroughness

The Dilemma

Modelers choose *speed* over standardization because comprehensive curation is too time-intensive

Our Solution

Template-Based LLM Workflow

Key Insight

LLMs can automate extraction **while enforcing standardization** through structured templates

Innovation 1: Templates

- YAML schemas define required metadata
- Enforce consistent structure
- Machine-parsable + human-readable
- Version controlled

Innovation 2: LLM Reasoning

- Complete transparency to derivation
- Captures interpretation decisions
- Documents study selection rationale
- Makes expert knowledge explicit

Goal

Achieve both **rigor** AND **efficiency** through automation

Six Core Metadata Dimensions

Our templates capture comprehensive parameter information:

1 Numerical values with uncertainty

- Point estimates, ranges, confidence intervals
- Statistical distributions (normal, lognormal, uniform)

2 Biological context

- Species, cell types, tissue compartments
- Disease state and cancer type

3 Experimental conditions

- Assay type, time points, treatments

4 Statistical distributions

- Distribution parameters and R bootstrap code

5 Literature provenance

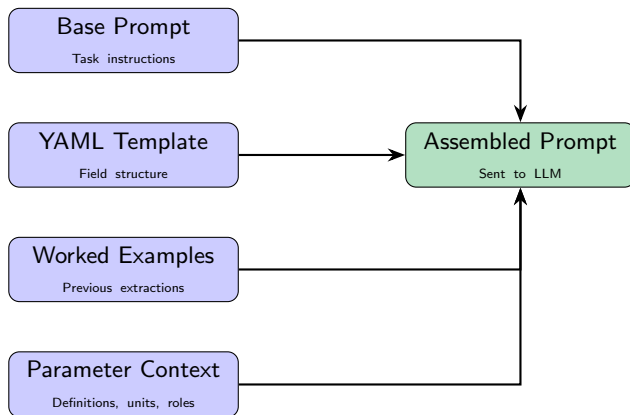
- DOI, figure/table numbers, specific passages

6 Cross-study pooling methodology

- Pooling strategy and heterogeneity assessment

System Architecture

Modular Prompt Assembly System



- **Modular** components, **configurable** via YAML
- **Reusable** across workflows, **maintainable** centrally

Three Workflow Types

1. Parameter Extraction

Purpose: Detailed metadata with full provenance

Output: All six metadata dimensions

Use case: Publication-ready parameter curation

2. Quick Estimates

Purpose: Rapid parameter initialization

Output: Ballpark values and ranges

Use case: Early-stage model testing

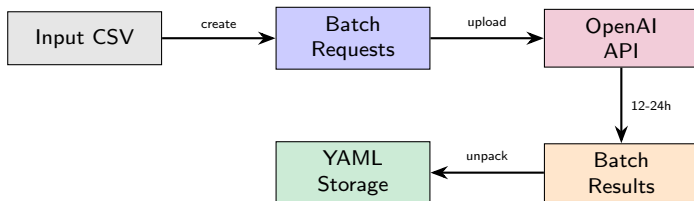
3. Test Statistics

Purpose: Validation targets from experimental data

Output: Statistical distributions with R bootstrap code

Use case: Model validation and calibration

Data Flow: Literature → YAML



- **Batch API:** 50% cost savings, async processing (12-24h)
- **Output:** Version-controlled YAML files in flat structure
- **Naming:** {param}-{author}-{cancer}-{hash}.yaml
 - Hash computed from study context enables multiple extractions per parameter

Example: YAML Output

```
parameter_name: "ki67_tumor_proliferation_rate"
canonical_scale: "log"
parameter_estimate:
  value: 0.0048
  unit: "1/h"
  type: "lognormal"
  confidence_interval: [0.0032, 0.0067]
biological_context:
  species: "human"
  cell_type: "PDAC_tumor_cells"
  disease_state: "pancreatic_ductal_adenocarcinoma"
experimental_design:
  assay_type: "Ki-67_immunohistochemistry"
  measurement_type: "proliferation_index"
  sample_size: 156
statistical_distribution:
  distribution_family: "lognormal"
  parameters: {meanlog: -5.34, sdlog: 0.18}
  r_bootstrap_code: |
    rlnorm(n, meanlog=-5.34, sdlog=0.18)
sources:
  - citation: "Smith_et_al.(2020)"
    doi: "10.1234/example"
    figure_or_table: "Figure_3B"
    specific_passage: "Median_Ki-67_index_was_48%..."
```

Complete, standardized, machine-parsable metadata!

Current Progress & Validation Plan

PDAC Application: Prototype Status

Model Context

Pancreatic ductal adenocarcinoma (PDAC) tumor microenvironment model
Therapeutic targets: checkpoint inhibitors, chemotherapy combinations

Current Status: Prototype Workflow

~**20 parameters** trialed through complete workflow
All core components functional, **but validation still needed**

What's Working:

- Template-guided extraction producing structured YAML
- Modular prompt assembly system
- Two-stage LLM review (extraction + automated validation)
- Batch processing with 12-24h turnaround
- Flat-file storage with version control

Key Resource: Legacy Parameter Database

Existing Asset

Extensive database of manually-curated parameters from prior QSP models

Format: Legacy YAML files with parameter values and metadata

Primary use for this study:

- **Validation ground truth:** Test LLM extraction accuracy at scale (50-100+ parameters)
- No new expert curation required

Secondary benefits:

- Contextualize PDAC parameters within broader QSP landscape
- Identify parameter patterns and coverage gaps across models

Key advantage

Large-scale validation without the time/cost of new expert curation

Primary Validation: Database Comparison

Approach: Compare LLM-extracted PDAC parameters to legacy manual curations

Validation metrics:

- Value agreement (correlation, fold-difference)
- Uncertainty quantification consistency
- Metadata completeness comparison
- Extraction accuracy rate

Scale: 50-100+ overlapping parameters

Bonus insights from database comparison:

- PDAC parameter landscape contextualized with other models
- Cross-model parameter variability characterization
- Coverage gap identification

Goal

Demonstrate LLM workflow produces accurate, reliable parameter metadata

Complementary Validation Options

Additional validation to strengthen the study (optional):

Option A: Expert Comparison for Gap Parameters

Use case: Parameters NOT in legacy database

Approach: Expert manually curates 10-20 parameters using identical sources

Benefit: Fills database coverage gaps + time savings measurement

Effort: High (requires expert time)

Option B: Downstream Model Performance

Use case: If PDAC model ready for testing

Approach: Use LLM parameters in model, compare predictions to data

Benefit: Real-world utility demonstration

Effort: Medium (depends on model readiness)

Implemented: Two-Stage LLM Review

Stage 1: Extraction

- Reads literature
- Fills YAML template
- Provides reasoning trace

Stage 2: Review

- 54-item checklist
- Citation verification
- Hallucination detection
- Returns corrected JSON

Checklist covers:

Parameter accuracy, statistical rigor, experimental design, source validation, formatting standards

Goal

Reduce expert burden while catching errors before manual validation

Paper Collaboration

Current Status

Completed

- ✓ Core workflows implemented
- ✓ Prototype tested (20 parameters)
- ✓ YAML templates finalized (v2)
- ✓ LLM review system (54-item)
- ✓ Flat-file storage system
- ✓ Paper outline drafted
- ✓ Onboarding materials prepared

Remaining

- ☐ Finalize validation approach
- ☐ Database comparison validation analysis
- ☐ Complementary validation (if needed)
- ☐ Generate figures/tables
- ☐ Write manuscript
- ☐ Supplementary materials
- ☐ Code release prep

Repository Organization

This repository: `qsp-llm-workflows`

General-purpose LLM workflow tools for parameter extraction

Scope: Reusable across any QSP model or disease area

Paper repository: `qsp-llm-workflows-paper` (to be created)

Paper-specific code, validation analyses, and figures

Scope: Validation study, results analysis, manuscript figures

Benefits:

- Clean separation between general tools and paper-specific work
- This repo remains focused and reusable
- Paper repo contains reproducible analyses for publication

What We Need: Division of Labor

Areas for collaboration (to discuss):

1 Database Comparison Validation

- Align LLM extractions with legacy YAML parameters
- Compute validation metrics (correlation, agreement, metadata quality)
- Statistical analysis scripting
- Contextualization: PDAC parameters vs. other models (secondary analysis)

2 Complementary Validation (if needed)

- Expert curation for gap parameters
- Time tracking and efficiency metrics

3 Writing & Figures

- Methods (workflow, templates, validation approach)
- Results (validation metrics, contextualized findings)
- Discussion (workflow accuracy, reproducibility, generalizability)
- Figures: validation plots, metadata quality comparisons

4 Supplementary Materials

- Complete template specifications
- LLM reviewer documentation
- Code repository preparation

Proposed Timeline

Timeframe	Milestones
Week 1-3	Database comparison validation Align extractions, compute metrics, contextualize with other models
Week 4-5	Complementary validation (if needed) Gap parameter expert curation, time tracking
Week 6-7	Figure generation and results analysis Validation plots, metadata quality comparisons
Week 8-9	Manuscript drafting All sections, supplementary materials
Week 10	Revisions and submission prep Internal review, final polishing

Target journal: *CPT: Pharmacometrics & Systems Pharmacology*

Timeline is preliminary and subject to discussion

Next Steps

Getting Started Today

Environment setup (5 minutes):

- 1 Activate virtual environment: `source venv/bin/activate`
- 2 Add API key to `.env`: `OPENAI_API_KEY=sk-...`
- 3 Clone sister repo: `qsp-metadata-storage`

Explore the codebase:

- Read: `docs/COLLABORATOR_ONBOARDING.md` (comprehensive guide)
- Review: `scratch/paper_outline_standardization.md`
- Browse: Existing extractions in `qsp-metadata-storage/`
- Try: Run a quick estimate workflow on a test parameter

Key documentation:

- `CLAUDE.md`: Complete codebase reference
- `templates/*.yaml`: See our schemas
- `prompts/*.md`: See how we prompt LLMs

Immediate Action Items

This week:

- 1 ✓ Attend this presentation / discussion
- 2 ☐ Review paper outline and decide on roles
- 3 ☐ Set up development environment
- 4 ☐ Run a test extraction workflow
- 5 ☐ Identify questions and areas needing clarification

By next meeting:

- 1 ☐ Finalize validation study design
- 2 ☐ Assign paper sections and figures
- 3 ☐ Create task tracking system (GitHub issues/project)
- 4 ☐ Schedule regular collaboration meetings

Let's discuss roles and next steps!

Thank You!

Questions & Discussion

*Onboarding guide & paper outline
(shared via email)*